

Name: _____

Quiz 1

Information Systems

3 September 2026

Use the following dataframe for all questions. Assume it has been assigned to `sales_df`. `index` is the data frame's **Index**, not a column named "Index".

index	Product	Category	Region	Sales	Units	Rating
1	Mouse	Electronics	West	80	10	4.1
2	Desk	Furniture	East	450	2	4.0
3	Chair	Furniture	South	275	5	3.8
4	Monitor	Electronics	West	600	4	4.3
5	Keyboard	Electronics	East	150	6	4.2
6	Lamp	Furniture	South	90	8	3.9
7	Tablet	Electronics	North	800	3	4.6
8	Bookshelf	Furniture	North	325	2	4.1
9	Webcam	Electronics	South	110	7	3.7
10	Laptop	Electronics	East	1200	3	4.5

- Which statement best describes the relationship between a pandas Series and a pandas DataFrame?
 - A Series can contain multiple columns, while a DataFrame can contain only one column.
 - A Series is one-dimensional, while a DataFrame is two-dimensional and can contain multiple Series as columns.
 - A Series and a DataFrame are both two-dimensional, but a DataFrame allows different data types in each row.
 - A DataFrame is simply a Series with an index removed.
- Which expression returns a dataframe containing only the **Product** and **Sales** columns?
 - `sales_df['Product', 'Sales']`
 - `sales_df['Product']['Sales']`
 - `sales_df[['Product'], ['Sales']]`
 - `sales_df[['Product', 'Sales']]`
- Which expression returns all rows where **Sales** is greater than 500?
 - `sales_df['Sales' > 500]`
 - `sales_df[sales_df > 500]`
 - `sales_df[sales_df['Sales'] > 500]`
 - `sales_df.loc['Sales' > 500]`

4. What does the following do: `sales_df.iloc[2]` ?
- (a) Selects the row whose index label is 2
 - (b) Selects the second row
 - (c) Selects the third row
 - (d) Selects the third column
5. Which expression selects the third row?
- (a) `sales_df[3]`
 - (b) `sales_df[2]`
 - (c) `sales_df[3:4]`
 - (d) `sales_df.iloc[2]`
6. What are the data types of: `sales_df['Sales']` and `sales_df[['Product','Rating']]`, respectively?
- (a) Index and DataFrame
 - (b) Series and DataFrame
 - (c) DataFrame and DataFrame
 - (d) Series and Series
7. What does the following expression return `sales_df['Units'].median()` ?
- (a) 4.5
 - (b) 6
 - (c) 4
 - (d) The number of rows containing Units values
8. Which code sorts the dataframe from highest Sales to lowest Sales?
- (a) `sales_df.sort_values('Sales', ascending=True)`
 - (b) `sales_df['Sales'].sort_values(ascending=False)`
 - (c) `sales_df.sort_values('Sales', ascending=False)`
 - (d) `sales_df.sort_values('Sales', descending=True)`
9. Which expression returns the Product name of the product with the highest Sales?
- (a) `sales_df.loc[sales_df['Sales'].max(), 'Product']`
 - (b) `sales_df[sales_df['Sales'] == sales_df['Sales'].max()][ 'Product']`
 - (c) `sales_df['Sales'].max()[ 'Product']`
 - (d) `sales_df[sales_df['Product'] == sales_df['Sales'].max()][ 'Product']`
10. What does the following return?
- ```
sales_df[sales_df['Units'] >= 5]['Product']
```
- (a) The Product values for rows having at least 5 units
  - (b) The Units values for products having at least 5 units
  - (c) All Product values sorted by Units
  - (d) The Product values for rows having fewer than 5 units

Answer Key

FIX

| Question | Answer |
|----------|--------|
| 1        | B      |
| 2        | C      |
| 3        | D      |
| 4        | A      |
| 5        | C      |
| 6        | C      |
| 7        | D      |
| 8        | B      |
| 9        | C      |
| 10       | D      |
| 11       | A      |
| 12       | D      |
| 13       | B      |